Exercise J-M

1. The number of real roots of the equation $\tan^{-1}\sqrt{x(x+1)}+\sin^{-1}\sqrt{x^2+x+1}=\frac{\pi}{4}$ is : **[JEE(Main)July2021]**

समीकरण $\tan^{-1}\sqrt{x(x+1)}+\sin^{-1}\sqrt{x^2+x+1}=\frac{\pi}{4}$ के वास्तविक मूलों की संख्या है : **[JEE(Main)July2021]**

(A) 1 (B) 4 (C) 0 (D) 2

Ans. C

Sol. Check domain of LHS

$x(x+1)\geq 0 \qquad 0\leq x^2+x+1\leq 1$

$x(x+1)\geq 0 \qquad 0\leq x^2+x+1\leq 1$

$\Rightarrow x^2+x=0$

$x=0,-1$

Both values of x do not satisfy the given equation.

Ans = 0

2. The value of $\tan\left(2\tan^{-1}\left(\frac{3}{5}\right)+\sin^{-1}\left(\frac{5}{13}\right)\right)$ is equal to : **[JEE(Main)July2021]**

$\tan\left(2\tan^{-1}\left(\frac{3}{5}\right)+\sin^{-1}\left(\frac{5}{13}\right)\right)$ का मान बराबर है : **[JEE(Main)July2021]**

(A) $\frac{151}{63}$ (B) $\frac{-291}{76}$ (C) $\frac{-181}{69}$ (D) $\frac{220}{21}$

Ans. D

Sol. $2\tan^{-1}\left(\frac{3}{5}\right)=\tan^{-1}\frac{\frac{3}{5}+\frac{3}{5}}{1-\frac{9}{25}}$

$=\tan^{-1}\left(\frac{15}{8}\right)$

$=\tan\left(\tan^{-1}\frac{15}{8}+\tan^{-1}\frac{5}{12}\right)$

$$= \tan\left(\tan^{-1}\frac{\frac{15}{8}+\frac{5}{12}}{1-\frac{75}{96}}\right)$$

$$= \frac{220}{21}$$

3. If the domain of the function $f(x) = \dfrac{\cos^{-1}\sqrt{x^2-x+1}}{\sqrt{\sin^{-1}\left(\dfrac{2x-1}{2}\right)}}$ is the interval $(\alpha, \beta]$, then $\alpha + \beta$ is equal to :

[JEE(Main)July2021]

यदि फलन, $f(x) = \dfrac{\cos^{-1}\sqrt{x^2-x+1}}{\sqrt{\sin^{-1}\left(\dfrac{2x-1}{2}\right)}}$ का प्रान्त, अन्तराल $(\alpha, \beta]$ है, तो $\alpha + \beta$ बराबर है :

[JEE(Main)July2021]

(A) 3/2 (B) 1/2 (C) 2 (D) 1

Ans. A

Sol. here

$0 \le x^2 - x + 1 \le 1$

Which is

$x^2 - x + 1 \ge 0$ and $x^2 - x + 1 \le 1$

$x \in R$ ....(1) $x^2 - x \le 0$

$x \in [0, 1]$ ....(2)

by (1) and (2)

$x \in [0, 1]$ ....(3)

Also

$$0 < \sin^{-1}\left(\frac{2x-1}{2}\right) \le \frac{\pi}{2}$$

$$0 < \frac{2x-1}{2} \le 1$$

$$0 < 2x - 1 \le 2$$

$$1 < 2x \le 3$$

$$\frac{1}{2} < x \le \frac{3}{2} \quad ....(4)$$

hence final answer is intersection of (3) and (4)

$x \in \left(\frac{1}{2}, 1\right]$

$\alpha = \frac{1}{2}$

$\beta = 1$

$\alpha + \beta = 3/2$

4. The domain of the function $\text{cosec}^{-1}\left(\frac{1+x}{x}\right)$ is : **[JEE(Main)Aug.2021]**

फलन $\text{cosec}^{-1}\left(\frac{1+x}{x}\right)$ का प्रांत है : **[JEE(Main)Aug.2021]**

(A) $x \in \left[-\frac{1}{2}, \infty\right) - \{0\}$ (B) $\left[-\frac{1}{2}, 0\right) \cup [0, \infty)$ (C) $\left[-\frac{1}{2}, 0\right) \cup [1, \infty)$ (D) $\left(-\frac{1}{2}, \infty\right) - \{0\}$

Ans. A

Sol. $\frac{1+x}{x} \le -1$ or $\frac{1+x}{x} \ge 1$

$\frac{1+2x}{x} \le 0$ or $\frac{1}{x} \ge 1$

$x \in \left[\frac{-1}{2} 0\right)$ or $x \in (0, \infty)$

$x \in \left[-\frac{1}{2}, \infty\right) - \{0\}$

5. If $\sum_{r=1}^{50} \tan^{-1} \frac{1}{2r^2} = p$, then the value of tan p is : **[JEE(Main)Aug.2021]**

यदि $\sum_{r=1}^{50} \tan^{-1} \frac{1}{2r^2} = p$ है, तो tan p का मान है : **[JEE(Main)Aug.2021]**

(A) $\frac{50}{51}$ (B) 100 (C) $\frac{101}{102}$ (D) $\frac{51}{50}$

Ans. A

Sol. $\because \tan^{-1}\frac{1}{2r^2} = \tan^{-1}\left(\frac{2}{1+\left(4r^2-1\right)}\right)$

$= \tan^{-1}\left(\frac{(2r+1)-(2r-1)}{1+(2r+1)(2r-1)}\right)$

$= \tan^{-1}(2r+1) - \tan^{-1}(2r-1)$

$\therefore \sum_{r=1}^{50} \tan^{-1}\left(\frac{1}{2r^2}\right) = \sum_{r=1}^{50}\left(\tan^{-1}(2r+1) - \tan^{-1}(2r-1)\right)$

$\therefore p = \tan^{-1}(101) - \tan^{-1}1$

$= \tan^{-1}\left(\frac{101-1}{1+101\cdot 1}\right)$

$\therefore \tan p = \frac{50}{51}$

6. If $\left(\sin^{-1}x\right)^2 - \left(\cos^{-1}x\right)^2 = a; 0 < x < 1, a \neq 0,$ then the value of $2x^2 - 1$ is : **[JEE(Main)Aug.2021]**

यदि $\left(\sin^{-1}x\right)^2 - \left(\cos^{-1}x\right)^2 = a; 0 < x < 1, a \neq 0$ है, तो $2x^2 - 1$ का मान है : **[JEE(Main)Aug.2021]**

(A) $\sin\left(\frac{4a}{\pi}\right)$ (B) $\cos\left(\frac{2a}{\pi}\right)$ (C) $\cos\left(\frac{4a}{\pi}\right)$ (D) $\sin\left(\frac{2a}{\pi}\right)$

Ans. D

Sol. $(\sin^{-1}x + \cos^{-1}x)(\sin^{-1}x - \cos^{-1}x) = a$

$\Rightarrow \frac{\pi}{2}\left(\frac{\pi}{2} - 2\cos^{-1}x\right) = a$

$\Rightarrow 2\cos^{-1}x = \frac{\pi}{2} - \frac{2a}{\pi}$

$\Rightarrow \cos^{-1}(2x^2-1) = \frac{\pi}{2} - \frac{2a}{\pi}$

$\Rightarrow 2x^2 - 1 = \cos\left(\frac{\pi}{2} - \frac{2a}{\pi}\right)$

$\Rightarrow 2x^2 - 1 = \sin\left(\frac{2a}{\pi}\right)$

7. Let M and m respectively be the maximum and minimum values of the function $f(x) = \tan^{-1}(\sin x + \cos x)$ in $\left[0, \frac{\pi}{2}\right]$.

Then the value of tan(M–m) is equal to : **[JEE(Main)Aug.2021]**

माना $\left[0, \frac{\pi}{2}\right]$ में फलन $f(x) = \tan^{-1}(\sin x + \cos x)$ के अधिकतम तथा न्यूनतम मान क्रमशः M तथा m हैं तो tan(M–m) का मान बराबर है : **[JEE(Main)Aug.2021]**

(A) $3+2\sqrt{2}$ (B) $2-\sqrt{3}$ (C) $3-2\sqrt{2}$ (D) $2+\sqrt{3}$

Ans. C

Sol. $y = \tan^{-1}\left(\sqrt{2}\sin\left(x+\frac{\pi}{4}\right)\right)$

$y = \tan^{-1}\left[\sqrt{2}\left[\frac{1}{\sqrt{2}}, 1\right]\right]$

$y = \tan^{-1}\left[1, \sqrt{2}\right]$

$y = \left[\frac{\pi}{4}, \tan^{-1}\sqrt{2}\right]$

Now $m = \frac{\pi}{4}$ $\qquad M = \tan^{-1}\sqrt{2} = \lambda \Rightarrow \tan\lambda = \sqrt{2}$

$= \tan(M-m)$

$= \tan\left(\lambda - \frac{\pi}{4}\right)$

$= \frac{\tan\lambda - 1}{\tan\lambda + 1}$

$= \frac{\sqrt{2}-1}{\sqrt{2}+1} \times \frac{\sqrt{2}-1}{\sqrt{2}-1}$

$= \left(\sqrt{2}-1\right)^2$

$= 2+1-2\sqrt{2}$

$= 3-2\sqrt{2}$

8. The domain of the function $f(x)=\sin^{-1}\left(\frac{3x^2+x-1}{(x-1)^2}\right)+\cos^{-1}\left(\frac{x-1}{x+1}\right)$ is : **[JEE(Main)Aug.2021]**

फलन $f(x)=\sin^{-1}\left(\frac{3x^2+x-1}{(x-1)^2}\right)+\cos^{-1}\left(\frac{x-1}{x+1}\right)$ का प्रांत है : **[JEE(Main)Aug.2021]**

(A) $\left[0,\frac{1}{4}\right]$ (B) $\left[\frac{1}{4},\frac{1}{2}\right]\cup\{0\}$ (C) $\left[0,\frac{1}{2}\right]$ (D) $[-2,0]\cup\left[\frac{1}{4},\frac{1}{2}\right]$

Ans. B

Sol. $f(x)=\sin^{-1}\left(\frac{3x^2+x-1}{(x-1)^2}\right)+\cos^{-1}\left(\frac{x-1}{x+1}\right)$

$-1\le\frac{3x^2+x-1}{x^2-2x+1}\le 1 \qquad -1\le\frac{x-1}{x+1}\le 1$

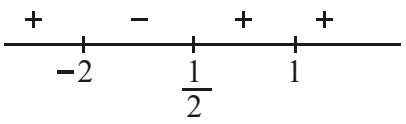
$\frac{3x^2+x-1}{x^2-2x+1}\ge -1$

$\frac{4x^2-x}{(x-1)^2}\ge 0\Rightarrow\frac{x(4x-1)}{(x-1)^2}\ge 0$

$x\in(-\infty,0]\cup\left[\frac{1}{4},1\right)\cup(1_1+\infty)$

$\frac{3x^2+x-1}{x^2-2x+1}-1\le 0$

$\frac{(x+2)(2x-1)}{(x-1)^2}\le 0$

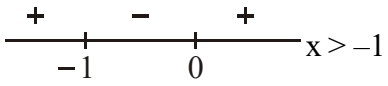


$x\in\left[-2,\frac{1}{2}\right]$

$x\in[-2,0]\cup\left[\frac{1}{4},\frac{1}{2}\right]$

$\frac{x-1}{x+1}\ge -1 \qquad \frac{x-1}{x+1}\le 1$

$\frac{x}{x+1} \geq 0$ $\qquad$ $\frac{1}{x+1} \geq 0$



$x \in (-\infty, -1) \cup [0, +\infty)$

$x \in [0, +\infty)$

$x \in \left[\frac{1}{4}, \frac{1}{2}\right] \cup \{0\}$

9. $\cos^{-1}(\cos(-5)) + \sin^{-1}(\sin(6)) - \tan^{-1}(\tan(12))$ is equal to : (The inverse trigonometric functions take the principal values) **[JEE(Main)Aug.2021]**

$\cos^{-1}(\cos(-5)) + \sin^{-1}(\sin(6)) - \tan^{-1}(\tan(12))$ बराबर है : (प्रतिलोम त्रिकोणमितीय फलन मुख्य मान लेते हैं ।) **[JEE(Main)Aug.2021]**

(A) $3\pi + 1$ (B) $4\pi - 11$ (C) $4\pi - 9$ (D) $3\pi - 11$

Ans. B

Sol. $\cos^{-1}(\cos(-5)) + \sin^{-1}(\sin 6) - \tan^{-1}(\tan 12)$

$= \cos^{-1}(\cos 5) + \sin^{-1}(\sin 6) - \tan^{-1}(\tan 12)$

$= (2\pi - 5) + 6 - 2\pi - (12 - 4\pi)$

$= 2\pi - 5 + 6 - 2\pi - 12 + 4\pi$

$= 4\pi - 11$

10. The set of all values of k for which $\left(\tan^{-1} x\right)^3 + \left(\cot^{-1} x\right)^3 = k\pi^3, x \in R$ has a solution in the interval : **[JEE(Main)June2022]**

k के सभी मानों, जिनके लिए $\left(\tan^{-1} x\right)^3 + \left(\cot^{-1} x\right)^3 = k\pi^3, x \in R$ है, का हल कौनसे अंतराल में है : **[JEE(Main)June2022]**

(A) $\left[\frac{1}{32}, \frac{7}{8}\right)$ (B) $\left(\frac{1}{24}, \frac{13}{16}\right)$ (C) $\left[\frac{1}{48}, \frac{13}{16}\right]$ (D) $\left[\frac{1}{32}, \frac{9}{8}\right)$

Ans. A

Sol. Note : Language is incomplete

$\left(\tan^{-1}(x)\right)^3 + \left(\cot^{-1}(x)\right)^3 = k\pi^3$ has a solution

$\Rightarrow k\pi^3$ will lie in range of LHS

$y = (\tan^{-1}(x))^3 + (\cot^{-1}(x))^3$

put $\tan^{-1}(x) = \alpha$, $\cot^{-1}(x) = \beta$

$\alpha+\beta=\frac{\pi}{2}$

$\alpha\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

$y=\alpha^3+\beta^3$

$=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$

$y=\frac{\pi^3}{8}-\frac{3\pi}{2}\alpha\left(\frac{\pi}{2}-\alpha\right),\ \alpha\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

$y)_{min}=\frac{\pi^3}{32}$ when $\alpha=\frac{\pi}{4}$

$y)_{max}\to\frac{7\pi^3}{8}$ when $\alpha\to-\frac{\pi}{2}$

$y\in\left[\frac{\pi^3}{32},\frac{7\pi^3}{8}\right)$

$\Rightarrow k\in\left[\frac{1}{32},\frac{7}{8}\right)$

11. The domain of the function $f(x)=\frac{\cos^{-1}\left(\frac{x^2-5x+6}{x^2-9}\right)}{\log_e\left(x^2-3x+2\right)}$ is : **[JEE(Main)June2022]**

फलन $f(x)=\frac{\cos^{-1}\left(\frac{x^2-5x+6}{x^2-9}\right)}{\log_e\left(x^2-3x+2\right)}$ का प्रांत है : **[JEE(Main)June2022]**

(A) $(-\infty,1)\cup(2,\infty)$ (B) $(2,\infty)$

(C) $\left[-\frac{1}{2},1\right)\cup(2,\infty)$ (D) $\left[-\frac{1}{2},1\right)\cup(2,\infty)-\left\{\frac{3+\sqrt{5}}{2},\frac{3-\sqrt{5}}{2}\right\}$

Ans. Bonus

Sol. $-1\le\frac{x^2-5x+6}{x^2-9}\le1$

$x^2-3x+2>0$

$x^2-3x+2\neq1$

$\Rightarrow x\in\left[-\frac{1}{2},1\right)\cup(2,\infty)-\left\{\frac{3+\sqrt{5}}{2},\frac{3-\sqrt{5}}{2},3\right\}$

Bonus

12. Let $x*y = x^2 + y^3$ and $(x*1)*1 = x*(1*1)$. Then a value of $2\sin^{-1}\left(\frac{x^4+x^2-2}{x^4+x^2+2}\right)$ is :

[JEE(Main)June2022]

माना $x*y = x^2 + y^3$ तथा $(x*1)*1 = x*(1*1)$ हैं। तो $2\sin^{-1}\left(\frac{x^4+x^2-2}{x^4+x^2+2}\right)$ का एक मान है :

[JEE(Main)June2022]

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{6}$

Ans. B

Sol. given $x * y = x^2 + y^3$

Now

$(x * 1) * 1 = x * (1 * 1)$

Using the given condition

$(x^2 + 1^3) * 1 = x * (1^2 + 1^3)$

$(x^2 + 1) * 1 = x * (2)$

$((x^2 + 1)^2 + 1) = x^2 + 8$

$x^4 + 2x^2 + 2 = x^2 + 8$

$x^4 + x^2 - 6 = 0$

$(x^2 + 3)(x^2 - 2) = 0$

$x^2 = 2$ is possible answer

Now

$$2\sin^{-1}\left(\frac{x^4+x^2-2}{x^4+x^2+2}\right)$$

$$2\sin^{-1}\left(\frac{4}{8}\right) = \frac{\pi}{3}$$

13. The value of $\tan^{-1}\left(\frac{\cos\left(\frac{15\pi}{4}\right)-1}{\sin\left(\frac{\pi}{4}\right)}\right)$ is equal to : **[JEE(Main)June2022]**

$\tan^{-1}\left(\frac{\cos\left(\frac{15\pi}{4}\right)-1}{\sin\left(\frac{\pi}{4}\right)}\right)$ का मान बराबर है : **[JEE(Main)June2022]**

(A) $-\frac{\pi}{4}$ (B) $-\frac{\pi}{8}$ (C) $-\frac{5\pi}{12}$ (D) $-\frac{4\pi}{9}$

Ans. B

Sol. $\tan^{-1}\left(\dfrac{\cos\left(4\pi-\dfrac{\pi}{4}\right)-1}{\sin\dfrac{\pi}{4}}\right)$

$-\tan^{-1}\left(\dfrac{1-\cos\dfrac{\pi}{4}}{\sin\dfrac{\pi}{4}}\right)$

$-\tan^{-1}\left(\tan\dfrac{\pi}{8}\right)=-\dfrac{\pi}{8}$

14. If the inverse trignometric functions take principal values, then $\cos^{-1}\left(\dfrac{3}{10}\cos\left(\tan^{-1}\left(\dfrac{4}{3}\right)\right)+\dfrac{2}{5}\sin\left(\tan^{-1}\left(\dfrac{4}{3}\right)\right)\right)$ is equal to : **[JEE(Main)June2022]**

यदि प्रतिलोम त्रिकोणमितीय फलन मुख्यमान लेते हैं, तो $\cos^{-1}\left(\dfrac{3}{10}\cos\left(\tan^{-1}\left(\dfrac{4}{3}\right)\right)+\dfrac{2}{5}\sin\left(\tan^{-1}\left(\dfrac{4}{3}\right)\right)\right)$ बराबर है : **[JEE(Main)June2022]**

(A) 0 (B) $\dfrac{\pi}{4}$ (C) $\dfrac{\pi}{3}$ (D) $\dfrac{\pi}{6}$

Ans. C

Sol. $\cos^{-1}\left(\dfrac{3}{10}\cos\left(\tan^{-1}\left(\dfrac{4}{3}\right)\right)+\dfrac{2}{5}\sin\left(\tan^{-1}\left(\dfrac{4}{3}\right)\right)\right)=0$

$\Rightarrow \tan\theta=\dfrac{4}{3}$

$\cos^{-1}\left(\dfrac{3}{10}\cos\theta+\dfrac{2}{5}\sin\theta\right)$

$\tan\theta=\dfrac{4}{3}=\dfrac{P}{B}$

$H=\sqrt{25}=25$

$\cos\theta=\dfrac{3}{5}$ $\sin\theta=\dfrac{4}{5}$

$\cos^{-1}\left(\dfrac{3}{10}\times\dfrac{3}{5}+\dfrac{2}{5}\times\dfrac{4}{5}\right)=\cos^{-1}\left(\dfrac{9}{50}+\dfrac{8}{25}\right)$

$$\cos^{-1}\left(\frac{25}{50}\right)=\cos^{-1}\left(\frac{1}{2}\right)=\frac{\pi}{3}$$

15. $\sin^{-1}\left(\sin\frac{2\pi}{3}\right)+\cos^{-1}\left(\cos\frac{7\pi}{6}\right)+\tan^{-1}\left(\tan\frac{3\pi}{4}\right)$ is equal to : **[JEE(Main)June2022]**

$\sin^{-1}\left(\sin\frac{2\pi}{3}\right)+\cos^{-1}\left(\cos\frac{7\pi}{6}\right)+\tan^{-1}\left(\tan\frac{3\pi}{4}\right)$ बराबर है : **[JEE(Main)June2022]**

(A) $\frac{11\pi}{12}$ (B) $\frac{17\pi}{12}$ (C) $\frac{31\pi}{12}$ (D) $-\frac{3\pi}{4}$

Ans. A

Sol. $\sin^{-1}\left(\sin\frac{2\pi}{3}\right)+\cos^{-1}\left(\cos\frac{7\pi}{6}\right)+\tan^{-1}\left(\tan\frac{3\pi}{4}\right)$

$\sin^{-1}\left(\sin\frac{\pi}{3}\right)+\cos^{-1}\left(\cos\left(\pi+\frac{\pi}{6}\right)\right)+\tan^{-1}\left(\tan\left(\pi-\frac{\pi}{4}\right)\right)$

$\sin^{-1}\left(\sin\frac{\pi}{3}\right)+\cos^{-1}\left(-\cos\frac{\pi}{6}\right)+\tan^{-1}\left(-\tan\frac{\pi}{4}\right)$

$\frac{\pi}{3}+\pi-\frac{\pi}{6}-\frac{\pi}{4}$

$=11\frac{\pi}{12}$

16. The value of $\cot\left(\sum_{n=1}^{50}\tan^{-1}\left(\frac{1}{1+n+n^2}\right)\right)$ is : **[JEE(Main)June2022]**

$\cot\left(\sum_{n=1}^{50}\tan^{-1}\left(\frac{1}{1+n+n^2}\right)\right)$ का मान है : **[JEE(Main)June2022]**

(A) $\frac{26}{25}$ (B) $\frac{25}{26}$ (C) $\frac{50}{51}$ (D) $\frac{52}{51}$

Ans. A

Sol. $\cot\sum\tan^{-1}\frac{(n+1)-n}{1+n(n+1)}$

$\cot\sum_{n=1}^{50}\tan^{-1}(n+1)-\tan^{-1}n$

$\cot\left(\tan^{-1}51-\tan^{-1}1\right)$

$\cot\tan^{-1}\frac{50}{52}=\frac{52}{50}=\frac{26}{25}$

17. The domain of the function $\cos^{-1}\left(\frac{2\sin^{-1}\left(\frac{1}{4x^2-1}\right)}{\pi}\right)$ is : **[JEE(Main)June2022]**

फलन $\cos^{-1}\left(\frac{2\sin^{-1}\left(\frac{1}{4x^2-1}\right)}{\pi}\right)$ का प्रांत है : **[JEE(Main)June2022]**

(A) $R-\left\{-\frac{1}{2},\frac{1}{2}\right\}$ (B) $(-\infty,-1]\cup[1,\infty)\cup\{0\}$

(C) $\left(-\infty,\frac{-1}{2}\right)\cup\left(\frac{1}{2},\infty\right)\cup\{0\}$ (D) $\left(-\infty,\frac{-1}{\sqrt{2}}\right]\cup\left[\frac{1}{\sqrt{2}},\infty\right)\cup\{0\}$

Ans. D

Sol. For domain $\cos^{-1}\left(\frac{2\sin^{-1}\left(\frac{1}{4x^2-1}\right)}{\pi}\right)$

$$-1\le\frac{2\sin^{-1}\left(\frac{1}{4x^2-1}\right)}{\pi}-\le1$$

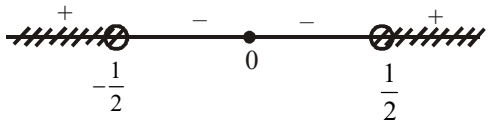
$$\Rightarrow\ -\frac{\pi}{2}\le\sin^{-1}\left(\frac{1}{4x^2-1}\right)\le\frac{\pi}{2}$$

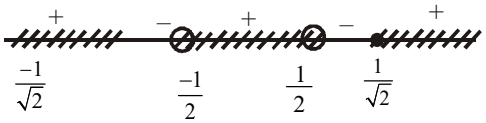
$$\Rightarrow\ -1\le\frac{1}{4x^2-1}\le1$$

$$\Rightarrow\ \frac{1}{4x^2-1}\le1\ \text{and}\ -1\le\frac{1}{4x^2-1}$$

$$\Rightarrow\ 0\le\frac{4x^2-1-1}{4x^2-1}\ \text{and}\ \ 0\le\frac{x^2}{(2x-1)(2x+1)}$$

$$\Rightarrow 0\le\frac{\left(\sqrt{2}x-1\right)\left(\sqrt{2}x+1\right)}{(2x-1)(2x+1)}\ \text{and}\ \le\frac{x^2}{(2x-1)(2x+1)}$$





$x \in \left(-\infty, \frac{-1}{\sqrt{2}}\right] \cup \left[\frac{1}{\sqrt{2}}, \infty\right) \cup \{0\}$

18. $50\tan\left(3\tan^{-1}\left(\frac{1}{2}\right)+2\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)\right)+4\sqrt{2}\tan\left(\frac{1}{2}\tan^{-1}\left(2\sqrt{2}\right)\right)$ is equal to _____.

[JEE(Main)June2022]

$50\tan\left(3\tan^{-1}\left(\frac{1}{2}\right)+2\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)\right)+4\sqrt{2}\tan\left(\frac{1}{2}\tan^{-1}\left(2\sqrt{2}\right)\right)$ का मान बराबर है _____

[JEE(Main)June2022]

Ans. 29

Sol. $50\tan\left(3\tan^{-1}\left(\frac{1}{2}\right)+2\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)\right)+4\sqrt{2}\tan\left(\frac{1}{2}\tan^{-1}\left(2\sqrt{2}\right)\right)$

$= 50\tan\left(\tan^{-1}\left(\frac{1}{2}\right)+2\tan^{-1}\left(\frac{1}{2}\right)+2\tan^{-1}(2)\right)+4\sqrt{2}\left(\frac{1}{\sqrt{2}}\right)$

$= 50\tan\left(\pi+\tan^{-1}\left(\frac{1}{2}\right)\right)+4$

$= 25 + 4$

$= 29$

19. $\tan\left(2\tan^{-1}\frac{1}{5}+\sec^{-1}\frac{\sqrt{5}}{2}+2\tan^{-1}\frac{1}{8}\right)$ is equal to : **[JEE(Main)July2022]**

$\tan\left(2\tan^{-1}\frac{1}{5}+\sec^{-1}\frac{\sqrt{5}}{2}+2\tan^{-1}\frac{1}{8}\right)$ बराबर है : **[JEE(Main)July2022]**

(A) 1 (B) 2 (C) $\frac{1}{4}$ (D) $\frac{5}{4}$

Ans. B

Sol. $\tan\left(2\left(\tan^{-1}\frac{1}{5}+\tan^{-1}\frac{1}{8}\right)+\tan^{-1}\left(\frac{1}{2}\right)\right)$

$= \tan\left[2\tan^{-1}\left(\frac{1}{3}\right)+\tan^{-1}\left(\frac{1}{2}\right)\right]$

$= 2$

20. If $0 < x < \frac{1}{\sqrt{2}}$ and $\frac{\sin^{-1} x}{\alpha} = \frac{\cos^{-1} x}{\beta}$, then a value of $\sin\left(\frac{2\pi\alpha}{\alpha+\beta}\right)$ is : **[JEE(Main)July2022]**

यदि $0 < x < \frac{1}{\sqrt{2}}$ तथा $\frac{\sin^{-1} x}{\alpha} = \frac{\cos^{-1} x}{\beta}$ हैं, तो $\sin\left(\frac{2\pi\alpha}{\alpha+\beta}\right)$ का एक समान है : **[JEE(Main)July2022]**

(A) $4\sqrt{\left(1-x^2\right)}\left(1-2x^2\right)$ (B) $4x\sqrt{\left(1-x^2\right)}\left(1-2x^2\right)$

(C) $2x\sqrt{\left(1-x^2\right)}\left(1-4x^2\right)$ (D) $4\sqrt{\left(1-x^2\right)}\left(1-4x^2\right)$

Ans. B

Sol. $\frac{\sin^{-1} x}{\alpha} = \frac{\cos^{-1} x}{\beta} = k$

$\sin^{-1} x = k\alpha$

$\cos^{-1} x = k\beta$

$k = \frac{\pi}{2(\alpha+\beta)}$ .....(i)

$\sin\left(\frac{2\pi\alpha}{\alpha+\beta}\right) = \sin\left(4\sin^{-1} x\right)$

$= 2\sin\left(2\sin^{-1} x\right)\cos\left(2\sin^{-1} x\right)$

$= 4x\sqrt{1-x^2}\left(1-2x^2\right)$

21. Considering only the principal values of the inverse trigonometric functions, the domain of the function $f(x) = \cos^{-1}\left(\frac{x^2-4x+2}{x^2+3}\right)$ is : **[JEE(Main)July2022]**

प्रतिलोम त्रिकोणमितीय फलन के केवल मुख्य मान लेते हुए, फलन $f(x) = \cos^{-1}\left(\frac{x^2-4x+2}{x^2+3}\right)$ का प्रांत है: **[JEE(Main)July2022]**

(A) $\left(-\infty, \frac{1}{4}\right]$ (B) $\left[-\frac{1}{4}, \infty\right)$ (C) $\left(-\frac{1}{3}, \infty\right)$ (D) $\left(-\infty, \frac{1}{3}\right]$

Ans. B

Sol. $\left|\frac{x^2+4x+2}{x^2+3}\right| \le 1$

$\Leftrightarrow \left(x^2-4x+2\right)^2 \le \left(x^2+3\right)^2$

$$\Leftrightarrow \left(x^2-4x+2\right)^2-\left(x^2+3\right)^2 \le 0$$

$$\Leftrightarrow \left(2x^2-4x+5\right)\left(-4x-1\right) \le 0$$

$$\Leftrightarrow -4x-1 \le 0 \rightarrow x \ge -\frac{1}{4}$$

22. Considering the principal values of the inverse trigonometric functions, the sum of all the solutions of the equation $\cos^{-1}(x) - 2\sin^{-1}(x) = \cos^{-1}(2x)$ is equal to : **[JEE(Main)July2022]**

प्रतिलोम त्रिकोणमितीय फलन के मुख्य मान लेते हुए समीकरण $\cos^{-1}(x) - 2\sin^{-1}(x) = \cos^{-1}(2x)$ के सभी हलों का योग है : **[JEE(Main)July2022]**

(A) 0 (B) 1 (C) $\frac{1}{2}$ (D) $-\frac{1}{2}$

Ans. A

Sol. $\cos^{-1} x = 2\sin^{-1} x = \cos^{-1} 2x$

$$\cos^{-1} x - 2\left(\frac{\pi}{2} - \cos^{-1} x\right) = \cos^{-1} 2x$$

$$\cos^{-1} x - \pi + 2\cos^{-1} x = \cos^{-1} 2x$$

$$3\cos^2 x = \pi + \cos^{-1} 2x \quad \text{....(1)}$$

$$\cos\left(3\cos^{-1} x\right) = \cos\left(\pi + \cos^{-1} 2x\right)$$

$$4x^3 - 3x = -2x$$

$$4x^3 = x \Rightarrow x = 0, \pm\frac{1}{2}$$

All satisfy the original equation

$$\text{sum} = -\frac{1}{2} \text{ to } +\frac{1}{2} = 0$$

23. The domain of the function $f(x) = \sin^{-1}\left(\frac{x^2-3x+2}{x^2+2x+7}\right)$ is : **[JEE(Main)July2022]**

फलन $f(x) = \sin^{-1}\left(\frac{x^2-3x+2}{x^2+2x+7}\right)$ का प्रांत है : **[JEE(Main)July2022]**

(A) $[1, \infty)$ (B) $[-1, 2]$ (C) $[-1, \infty)$ (D) $(-\infty, 2]$

Ans. C

Sol. $f(x) = \sin^{-1}\left(\frac{x^2-3x+2}{x^2+2x+7}\right)$ Domain

$\frac{x^2-3x+2}{x^2+2x+7} \geq -1$ and $\frac{x^2-3x+2}{x^2+2x+7} \leq 1$

$2x^2 - x + 9 \geq 0$ and $5x \geq -5 \Rightarrow x \geq -1$

$x \in R$

Hence Domain $x \in [-1, \infty)$

24. Let $x = \sin(2\tan^{-1}\alpha)$ and $y = \sin\left(\frac{1}{2}\tan^{-1}\frac{4}{3}\right)$. If $S = \{\alpha \in R : y^2 = 1 - x\}$, then $\sum_{\alpha \in S} 16\alpha^3$ is equal to _____.

[JEE(Main)July2022]

माना $x = \sin(2\tan^{-1}\alpha)$ तथा $y = \sin\left(\frac{1}{2}\tan^{-1}\frac{4}{3}\right)$ हैं। यदि $S = \{\alpha \in R : y^2 = 1 - x\}$ है, तो $\sum_{\alpha \in S} 16\alpha^3$ बराबर है _____

[JEE(Main)July2022]

Ans. 130

Sol. $\because x = \sin\left(2\tan^{-1}\alpha\right) = \frac{2\alpha}{1+\alpha^2}$ ...(i)

and $y = \sin\left(\frac{1}{2}\tan^{-1}\frac{4}{3}\right) = \sin\left(\sin^{-1}\frac{1}{\sqrt{5}}\right) = \frac{1}{\sqrt{5}}$

Now, $y2 = 1 - x$

$\frac{1}{5} = 1 - \frac{2\alpha}{1+\alpha^2}$

$\Rightarrow 1 + \alpha^2 = 5 + 5\alpha^2 - 10\alpha$

$\Rightarrow 2\alpha^2 - 5\alpha + 2 = 0$

$\therefore \alpha = 2, \frac{1}{2}$

$\therefore \sum_{\alpha \in S} 16\alpha^3 = 16 \times 2^3 + 16 \times \frac{1}{2^3}$

$= 130$

25. For $k \in R$, let the solutions of the equation $\cos(\sin^{-1}(x \cot(\tan^{-1}(\cos(\sin^{-1} x))))) = k$, $0 < |x| < \frac{1}{\sqrt{2}}$ be α and β, where the inverse trigonometric functions take only principal values. If the solutions of the equation $x^2 - bx - 5 = 0$ are $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ and $\frac{\alpha}{\beta}$, then $\frac{b}{k^2}$ is equal to _____. **[JEE(Main)July2022]**

माना $k \in R$ के लिए समीकरण $\cos(\sin^{-1}(x \cot(\tan^{-1}(\cos(\sin^{-1} x))))) = k$, $0 < |x| < \frac{1}{\sqrt{2}}$ के हल α तथा β हैं, जहाँ प्रतिलोम त्रिकोणमितीय फलन केवल मुख्य मान लेते हैं। यदि समीकरण $x^2 - bx - 5 = 0$ के हल $\frac{1}{\alpha^2}+\frac{1}{\beta^2}$ तथा $\frac{\alpha}{\beta}$ है, तो $\frac{b}{k^2}$ बराबर है _____ **[JEE(Main)July2022]**

Ans. 12

Sol. $\cos\left(\sin^{-1} x\right) = \cos\left(\cos^{-1}\sqrt{1-x^2}\right) = \sqrt{1-x^2}$

$$\cot\left(\tan^{-1} 1-x^2\right) = \cot\cot^{-1}\left(\sqrt{\frac{1}{\sqrt{1-x^2}}}\right) = \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow \cos\left(\sin^{-1}\left(\frac{x}{1-x^2}\right)\right) = \frac{\sqrt{1-2x^2}}{\sqrt{1-x^2}}$$

$$\Rightarrow \frac{\sqrt{1-2x^2}}{\sqrt{1-x^2}} = k$$

$$\Rightarrow 1-2x^2 = k^2\left(1-x^2\right)$$

$$\Rightarrow \left(k^2-2\right)x^2 = k^2-1$$

$$x^2 = \frac{k^2-1}{k^2-2}$$

$$\alpha = \sqrt{\frac{k^2-1}{k^2-2}} \Rightarrow \alpha = \frac{k^2-1}{k^2-2}$$

$$\beta = \sqrt{\frac{k^2-1}{k^2-2}} \Rightarrow \beta^2 = \frac{k^2-1}{k^2-2}$$

$$\frac{1}{\alpha^2}+\frac{1}{\beta^2} = 2\left(\frac{k^2-2}{k^2-1}\right) \& \frac{\alpha}{\beta} = -1$$

$$\text{Sum of roots} = \frac{1}{\alpha^2}+\frac{1}{\beta^2}+\frac{\alpha}{\beta} = b$$

$$\Rightarrow \frac{2\left(k^2-2\right)}{k^2-1}-1 = b \quad .....(1)$$

$$\text{Products of roots} = \left(\frac{1}{\alpha^2}+\frac{1}{\beta^2}\right)\frac{\alpha}{\beta} = -5$$

$$\Rightarrow \frac{2\left(k^2-2\right)}{k^2-1}(-1)=-5$$

$$\Rightarrow 2k^2-4=5k^2-5$$

$$\Rightarrow 3k^2=1 \Rightarrow k^2=\frac{1}{3} \text{ .....Putin (1)}$$

$$\Rightarrow b=\frac{2\left(k^2-2\right)}{k^2-1}-1=5-1=4$$

$$\frac{b}{k^2}=\frac{4}{\frac{1}{3}}=12$$